UNIVERSITY OF GREENWICH
FACULTY OF ENGINEERING AND SCIENCE
SCHOOL OF COMPUTING AND MATHEMATICAL SCIENCES

IMAGE PRIVACY LEAKAGE IN AI: A STUDY OF GRADIENT RECONSTRUCTION AND FACE PIXELATION

AGSAR YASIN
(001328612)

Supervisor: Muhammad Waqas

Word count: 16508

COMP1682 Final Year Individual Project

A dissertation submitted in partial fulfilment of the requirements for the degree of

BSc (Hons) COMPUTER SCIENCE IN CYBERSECURITY

April 2026

ABSTRACT

This study is research-based. This research study and explore privacy leakage in image based artificial intelligence systems, focusing on gradient leakage and model inversion attacks. As people upload a personal image to AI platform, there is a growing concern that these image may be stored and processed or used in the model training. Creating privacy risk of users. Even when the raw images are not directly shared private raw information may still leak through gradients. This project studies these risk and reviews existing defence methods, including differential privacy and dropout l2 regulation. And propose defence idea developed by adapting an existing defence thinking called face pixelation during training to reduce identity-sensitive face details. To support the research, the project includes a small proof of concept demonstration with two models. A baseline model that reconstructs an approximate image from leaked gradients data, and defended model that uses pixelated for reconstruction and produces a blurry output, this study finds that image privacy leakage is a serious issue in AI systems and suggests that reducing facial detail before training can help lower reconstruction quality and reduce privacy risk . it also argues that company-side changes in data handling and image storage are important for better privacy protection

DECLARATION OF AI USE

Please complete this part when you have used AI during the process of undertaking this assignment to acknowledge the ways in which you have used it.

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – YES
- To create an outline of the topic – NO
- To explain concepts – YES
- To support my use of language – YES
- To summarise the following articles/resources: – YES
    1. AI tools were used occasionally throughout this project to support language improvement, sentence clarity, and general phrasing. They were also used at times to help explain ideas more clearly during the writing process. The research, analysis, implementation, and final decisions remained my own throughout.

ACKNOWLEDGEMENTS

I would like to thank my supervisor, Muhammad Waqas, and Ehsan Nowroozi, for their guidance, support, and feedback throughout this project. I am also grateful to my friends and colleagues for their helpful suggestions and encouragement during the development of the work. In addition, I would like to thank my brother, who works in the cyber cell for his advice and support. Their feedback and encouragement were very valuable throughout this project

CONTENTS

Abstract ii
Declaration of AI Use iii
Acknowledgements iv
Contents v
1 Introduction 1
1.1 Background 2
1.2 Aims and Objectives 4
1.3 Methodology 5
1.4 Report Roadmap 6
2 Literature Review 7
2.1 Introduction 7
2.2 Existing defence against privacy leakage 9
2.3 Supporting background defence 11
2.4 Conclusion 11
3 Product Review (if applicable) 13
3.1 Introduction 13
3.2 Product 1 Gradient inversion reconstruction approach 13
3.3 Product 2 Face de-identification / pixel-level defence approach 15
3.4 Key findings 16
3.5 Conclusion 17
4 Requirements Analysis and Design 18
4.1 Introduction 18
4.2 Requirements Analysis 18
4.3 Design 22
4.4 Conclusions 25
5 Implementation 26
5.1 Introduction 26
5.2 Code outline 26
5.3 Baseline pipeline implementation 26
5.4 Defended pipeline implementation 27
5.5 Testing 27
5.6 Conclusion 28
5.7 Introduction 30
5.8 Results 30

5.9 Evaluation 34
5.10 Conclusion 36
6 Legal, Social, Ethical and Professional Issues 37
7 Conclusion 38
7.1 Summary of the work 38
7.2 Overall outcomes 38
7.3 Limitations and reflection 38
7.4 Future Work 38
APPENDIX A - Project Proposal 43
a.1 Revised Proposal/Project Aim (if applicable) 43
a.2 Revised Project Plan 43
a.3 Original Proposal 43
a.4 Project Plan 50
APPENDIX B - Testing documentation 52
b.1 Testing plan 52
b.2 Baseline pipeline Testing 52
b.3 Defended pipeline testing 53
b.4 Repeated testing result 53
b.5 Supporting literature comparison 54

1 INTRODUCTION

Image generative AI platforms have become part of everyday life. People use the AI platforms to edit photographs, generate images, generate video content, and to edit their pictures according to the user's prompt. Behind these systems are machine learning models that are trained on large amounts of visual data. While these kinds of tools are always useful, they also create some important privacy concerns. Personal images can contain faces, home environments, documents, and other sensitive details that reveal far more than a user expects. Because of this image privacy has become a serious issue in modern AI research. Surveys of privacy attacks in machine learning and it shows that the models can leak information about the data used to train them, which means that privacy risks do not only come from direct Data breaches but also from the behaviour of the model itself. **(Rigaki and Garcia, 2023; Yang et al., 2025)**

This problem is mainly in company-based AI platforms. When users upload their images to online services, they may not know how their images are used, how they are stored, processed, or reused. In some cases, platforms clearly show that uploaded or backed-up content can remain within the company systems and later be reprocessed by automated features. For example, Google Photos can reappear as memories of a year through its memory feature. This doesn't prove that every uploaded image is always used by the training model, but it does show why users may be concerned about how their personal images are handled after uploading.

One of the main privacy risks studied in this project is **Gradient leakage**. In collaborative and **federal learning**, raw data is not shared directly. Instead of **gradients** or parameter updates being exchanged a model can learn from distributed data. This was originally seen as a more privacy-friendly approach because the original data stays local. However, research has shown that **gradients** can reveal sensitive information about the original input. Deep learning from the **gradients** demonstrated that private training data can be recovered from shared **gradients**, and later work showed that this problem remains serious even for deeper and more realistic architectures. (Zhu, Liu, and Han, 2019; Geiping et al., 2020)

A second privacy risk is **model inversion**; unlike gradient leakage, **model inversion** does not mainly depend on access to training **gradients**. Instead, it uses model outputs such as confidence values or predictions to assume sensitive information about the data on which the model was trained. Foundational work on this area showed that a **model inversion attack** could recover recognised face-related information from machine learning systems. This is important because it means privacy risk can appear not only during training, but also after deployment, when a trained model is made available for use.(Abadi et al., 2016; Hamida et al., 2023)

This project investigates privacy leakage in image-based artificial intelligence systems, with a particular focus on gradient leakage and model inversion attacks. It also studies defence mechanisms that may reduce the risk of exposing sensitive image information. Two existing defences are reviewed from the literature: **Differential privacy** and **Dropout with L2 regularisation**. **Differential privacy** is one of the most widely discussed privacy-preserving methods because it limits the influence of individual training samples, although it also introduces a trade-off between privacy and model utility. **Dropout with L2 regularisation** is a more general regularisation- based approach that helps reduce overfitting and memorisation, making leakage harder but without providing the same formal guarantees as **differential privacy.**

Alongside these existing defences, this project also proposes a simple privacy idea called **face pixelation** during training. The idea is that if identity-sensitive facial detail is reduced before training, then the gradients derived from that image should reveal less recognised information. This direction is supported by wider research on **face de-identification**, which shows that reducing visible facial identity information is an important privacy strategy in image-based systems.

This project is mainly research-based, but it is also supported by a **small proof-of-concept demonstration.** Two models are used. The first is a **baseline** leakage model, where an input image produces **leaked gradient** data that can be used to reconstruct and approximate an image. The second model is a **defended model**, where the image is pixelated before training, producing a blurrier and less accurate reconstruction. By comparing these two cases, the project aims to show the seriousness of image privacy leakage and the possible value of privacy-preserving defences in reducing the quality of leaked reconstructions.

.

1.1 Background

Artificial intelligence systems that process and generate images are used in many parts of everyday life, including social media, entertainment, healthcare, surveillance, and online creative platforms. These systems usually rely on machine learning models trained on very large amounts of visual data. While this has improved image generation, recognition, and analysis, it has also raised serious privacy concerns. Images can contain faces, private environments, documents, and other sensitive details, so they often reveal much more than a user may expect.

This issue becomes even more important in company-based AI platforms, where users may upload personal images without fully knowing how those images are stored, processed, retained, or potentially reused. Depending on platform policies and system design, uploaded images may contribute to future service improvement or broader model development. In practice, the same image could potentially

support different visual tasks, such as portrait-style learning, background human figures, or character-like image generation. This creates a privacy concern because users may not have full visibility into what happens to their image data once it has been uploaded. **(Rigaki and Garcia, 2023; Yang et al., 2025)**

A well-known example of this wider problem is **Clearview AI**. A legal analysis of the case explains that the company became widely known in January 2020 after reports revealed that it had scraped internet images to build a facial-recognition database containing around 3 billion pictures, and that by October 2021 this had reportedly increased to 10 billion pictures. The same source explains that scraped pictures, together with pictures uploaded into the application, were treated as inputs to the system and stored on the company's servers. This case is important because it shows how real people's face images can be collected on a very large scale and used in facial-recognition systems without their knowledge or meaningful consent. Dul (2022)

More recent AI security incidents show that the wider AI ecosystem can also expose sensitive information in other ways. In January 2025, Wiz research reported that it had identified a publicly accessible Deepseek clickhouse database that allowed access to internal data and contained over a million lines of log streams, including chat history, secret keys, backend details, and other highly sensitive information. This is important to present research because it shows that AI platforms can expose sensitive data through weak infrastructure security, which strengthens the wider concern that once data enters an AI system.

A similar concern appeared in the AI agent ecosystem. On 5 February 2026, Sync published research on **OpenClaw** and the ClawHub marketplace, reporting that its researchers scanned 3,984 skills and found 283 skills with critical flaws that exposed sensitive credentials. According to the report, some of these skills instructed AI agents to mishandle secrets by passing API keys, passwords, and even credit card numbers through the LLM context window and plaintext logs. This matters to present research because it shows that AI-related systems can mishandle sensitive data not only through direct breaches, but also through insecure design choices that expose private information indirectly.

The specific problem addressed in this project is **privacy leakage from image-based AI systems**, with particular focus on **gradient leakage** and **model inversion attacks**. Gradient leakage is important because many distributed or federated learning approaches assume that sharing gradients is safer than sharing raw data. However, research has shown that gradients can still reveal sensitive information about the original training inputs. *Deep Leakage from Gradients* demonstrated that private training data can be recovered from shared gradients, while later work showed that this problem remains serious even for deep and realistic architectures. More recent studies have continued to improve these attacks, showing that gradient inversion remains possible even in more advanced federated-learning and secure-aggregation settings.

A second important threat is **model inversion**. Unlike **gradient leakage**, model inversion attacks exploit model outputs, such as confidence values or predictions, to infer sensitive properties of training data or reconstruct recognisable information. Foundational work in this area showed that model inversion attacks could recover recognisable face-related information from machine learning systems. This demonstrates that privacy risks are not limited to the training stage, but can also continue after a model has been deployed. (Fredrikson, Jha, and Ristenpart, 2015; Yang et al., 2025)

This problem is particularly relevant to cybersecurity because it concerns data confidentiality, privacy protection, and the security of AI systems. It is also significant because several groups are affected by it. Users may lose control over personal image data, organisations may face legal and ethical consequences, and developers are increasingly expected to build systems that are secure by design. Even when a company claims that only internal systems, gradients, or model-related information were exposed and that no raw user photos were directly leaked, privacy research shows that this does not automatically mean users are safe. If gradients or related training information contain enough detail, an attacker may still reconstruct an approximate version of the original image. In that sense, indirect leakage can still become a real privacy breach.

Existing research has already proposed several ways to reduce these risks. Differential Privacy is one of the most established approaches because it limits the contribution of individual training samples, although it introduces a trade-off between privacy and model utility. Regularisation methods such as dropout and L2 regularisation can also reduce overfitting and memorisation, although they do not provide the same formal privacy guarantees. In addition, research on face de-identification suggests that reducing identity-sensitive facial detail is an important direction for protecting privacy in image-based systems. (Abadi et al., 2016; Hamida et al., 2023; Scheliga et al., 2025)

This project builds on that existing work by reviewing the privacy risks of gradient leakage and model inversion, examining two existing defences from the literature, **Differential Privacy** and **Dropout with L2 regularisation**, and proposing a task-specific defence idea called **Face Pixelation During Training**. The project also includes a small proof-of-concept comparison between a baseline leakage model and a defended model to show how reducing facial detail before training may lower the quality of reconstructed outputs. In this way, the background to the project is not only the broader privacy problem in AI, but also the need to better understand how image privacy can be protected in practice.

1.2 Aims and Objectives

Aims

This project aims to investigate privacy leakage in image-based artificial intelligence systems, with a particular focus on gradient leakage and model inversion attacks, and to examine how existing and proposed defence mechanisms can reduce the risk of reconstructing sensitive image information.

Objectives

- To review existing literature on privacy leakage in machine learning, with particular attention to image-based AI systems.
- To examine how a gradient leakage attack can expose private training data allow approximate image reconstruction.
- To analyse model inversion attacks and explain how they threaten sensitive image information in deployed AI systems.
- To investigate two existing defence mechanisms, Differential privacy and Dropout with L2 regularisation, and assess their strengths and limitations in reducing privacy leakage.
- To develop and discuss a proposed defence approach, face pixelation, during training, as a method for reducing identity-sensitive facial detail before training
- To implement a baseline proof-of-concept model that demonstrates image reconstruction from leaked gradients
- To implement a defended version of the model in which the image is pixelized before training and compare the reconstructed output with the baseline version
- To evaluate whether the proposed defence reduces the quality, accuracy and recognisability of reconstructed images.

1.3 Methodology

This project uses a **research-based approach** and is supported by a small **model for proof-of-concept.** This approach was chosen because the project is mostly about understanding a privacy issue, by looking at what previous research says about it and exploring possible defence methods, rather than making a full software product. Since the project focuses on privacy leakage based on image generative AI systems, it was important to use **peer-reviewed journals** and recent studies to build a reliable understanding of the topic.

The main method used in this project is literature analysis. This was suitable because the project looks at two privacy threats that have already been widely discussed in research: gradient leakage and model inversion. studying this literature helped explain how these attacks work, why they matter, and what defence methods have already been proposed. It also helped support the choice of **Differential privacy** and **Dropout L2 regularisation** as two existing defences examined in this study.

A small model was also developed. This was included because the privacy issue can be understood more clearly through a simple practical demonstration, not only through theory. For this reason, a **proof-of-**

concept system was built using Python to represent both a baseline leakage case and a **defended case**. The baseline model shows how an input image can produce gradients that allow an approximate reconstruction, while the defended model applies **face pixelation** during training so that the reconstructed result becomes blurrier and less accurate. This made it possible to demonstrate the privacy problem in a simple, controlled way without needing a large word system.
Overall, the methodology combines secondary research with practical demonstration. This literature provides the theoretical basis of the project, while the model gives a small technical example to support the discussion. Using both together makes it easier to draw a conclusion about how serious privacy leakage can be in the image generative AI systems and whether the selected defences reduce the impact of reconstruction attacks, even If they do not remove the risk completely.

1.4 Report Roadmap

This report starts by introducing the wider privacy problem in image-based artificial intelligence and explaining why it is worth studying. It then moves to the literature review, which looks at the main research on gradient leakage, model inversion, and privacy-preserving defences. This chapter gives the background needed for the rest of the project and shows how the study fits into existing research.

The report then moves on to the requirements and design of the project. This section explains the scope of the work, the assumptions made, and how the proof-of-concept system was planned. After that, the implementation chapter describes how the baseline leakage model and the defended model were developed, including how the proposed pixelation-based defence was applied.

The next chapter presents the results and evaluation. This section compares the outputs from both models to see whether the defended version reduces the quality or recognisability of reconstructed images compared with the baseline case. The report then discusses the legal, social, ethical, and professional issues linked to image privacy and AI systems. Finally, the conclusion summarises the main findings, explains the limitations of the project, and suggests possible future work.

2 LITERATURE REVIEW

2.1 Introduction

Privacy exposure in machine learning has been identified as a serious problem because it is possible for an attacker to learn information private to the learning process. This is particularly important when the machine learning algorithms are learning on images, in which case the information can be sensitive visual information other than behaviour. Zhu, Liu and Han show that publicly released gradients leak private training data and private labels, and pixels can be perfectly reconstructed. Geiping et al considerably improve these attacks, and show that high-resolution images can still be reconstructed from parameter gradients, and that averaging many images or many iterations of gradient descent does not guarantee privacy. This is at the heart of the problem for our setup: an opponent can reconstruct an approximate original image even when the raw image is never directly shared (Zhu, Liu and Han, 2019; Geiping et al., 2020).

This is not only an academic issue, since many AI systems require training on real individual photographs. Clearview AI was first brought to public attention by Dul in January 2020, when it was revealed that the company had scraped about 3 billion images from the internet. Later reports indicate that about 10 billion images were scraped. While the paper explains that it is possible for Clearview to return the images together with the website URLs only if an individual has uploaded a portrait to Clearview, this case nevertheless shows how images of real people can be collected, stored and used in AI systems at scale, often without the individuals depicted in the images having a meaningful awareness of or providing meaningful consent. (Dul, 2022).

In the context of the existing literature, Rigaki and Garcia review privacy attacks applicable to machine learning like reconstruction attacks, model inversion attacks, inference attacks and model extraction attacks as well as the properties associated with information leaks like memorisation, weak generalization and structural properties of the models. For this project, however, we mainly concentrate on gradient leakage as it is most closely related to our primary research question about whether the leaked gradients provide the image-reconstructor with sufficient information to reconstruct the image. (Rigaki and Garcia, 2023).

2.1.1 Gradient leakage as the main research focus

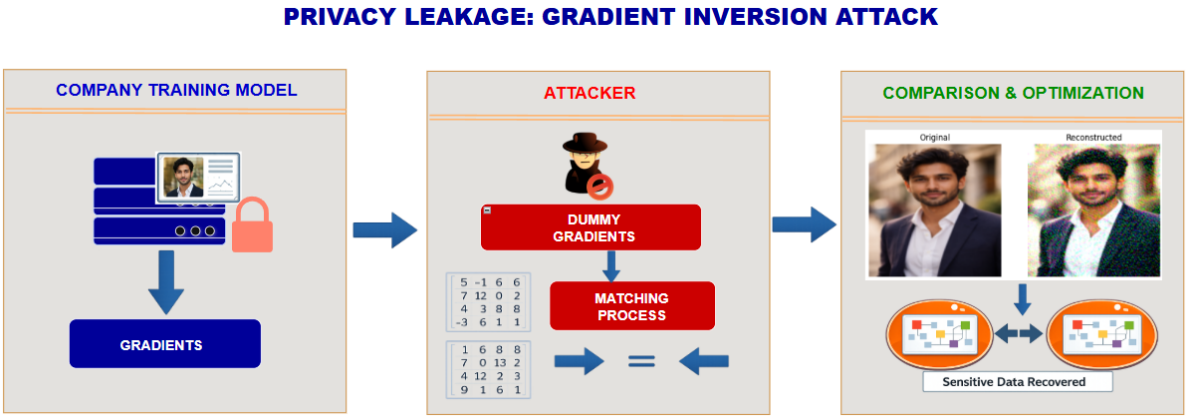


Fig 1.1 shows the real privacy

The main attack discussed in this paper is gradient leakage, which directly explains why reconstruction works. Zhu, Liu and Han show that one can recover the private training data from the released gradients by minimizing a function that reconstructs the training inputs and targets. The implications of these results are meaningful, as they disprove the assumption that gradients are benign and serve as the basis for the baseline leakage model adopted by this work, which describes the case when an image is used to compute gradients that the opponent seeks to reconstruct.

This later work shows the threat to be more real than imagined, with Geiping et al arguing that these previous attacks provided a false sense of security in toy domains. They also show that the attack can successfully reconstruct high-quality images from gradients in a realistic deep/trained architecture, and that averaging multiple local (unlabelled) examples does not guarantee that privacy is preserved. This is important not only because it shows that gradient leakage is not just a toy example, but because it is a more serious privacy violation in realistic learning scenarios, so we will use it as our primary attack model.

Subsequent work follows this line of reasoning, finding the original inversion attack does not scale for larger batch sizes and images, and then establishing a more general framework for image and label reconstruction under more complex conditions. These works show that gradients inversion is an open area of research, and that the project's defence will be evaluated against an improving series of adversarial attack techniques, rather than only the initial set of attacks.

The literature also suggests that leakage need not be limited to the image. Gat, Sharif, and Sundararajan show that labels can be extracted from bias gradients in federated learning. This is meaningful since reconstructive attacks often rely on correct label extraction. Yang et al. further show that, even with unbiased sampling-based secure aggregation, approximate gradients could be used to reconstruct images with pixel-wise accuracy. This is an important result for the current work as it extends the privacy threat: if useful gradient information is leaked, both the images and their corresponding labels can be reconstructed.

2.1.2 Model inversion as a related privacy threat

Although the focus of this project is gradient leakage, there is a related and equally important attack called Model Inversion, which shows that the attacker does not only need the gradients. Fredikson, Jha and Ristenpart show that the attacker can obtain sensitive features by model access alone, even without a training dataset. For example, given the name of the target person, recognizable images of their face can be recovered from a facial recognition model. This property is also problematic since it implies that image privacy can potentially be broken at inference time, in addition to during training time when gradients are shared. (Fredrikson, Jha and Ristenpart, 2015).

Thus, model inversion fits into the gradient leakage literature. Rigaki and Garcia placed it in the broader context of privacy attacks, arguing that machine learning privacy issues should be viewed as just one of a larger class of problems, rather than being an isolated issue. This is important for the project because it positions gradient leakage as the main implemented attack without denying that deployed models may also leak sensitive information through their outputs (Rigaki and Garcia, 2023).

2.2 Existing defence against privacy leakage

The literature shows that privacy defences in machine learning mainly work by making reconstruction harder, reducing the amount of useful information available to the attacker, or weakening the quality of what can be recovered. In this project, the important point is not simply that a defence exists, but how it affects gradient leakage attacks in practice. Existing work suggests that defences do not usually make leakage impossible in every case. Instead, they are better understood as methods that reduce the success, accuracy or usefulness of the attack.

2.2.1 Differential privacy

Differential Privacy is one of the strongest defences in the literature because it protects training data by limiting how much influence any single data sample has on the information released during learning.

Abadi et al. describe it as a method for training deep neural networks with privacy guarantees by controlling privacy loss within a formal framework. In practical terms, this affects gradient leakage attacks by making the shared training signal less directly tied to any one original training example. As a result, it becomes harder for an attacker to use gradients to recover the exact input image.

This matters for the project because gradient leakage attacks depend on gradients containing enough information to match back to the original input. If Differential Privacy reduces the amount of exact sample-specific information available in those gradients, then reconstruction quality should fall. The defence therefore works not by removing the attack itself, but by weakening the information the attack relies on.

The main drawback is it reduces the amount of useful information available to the attacker, but may also reduce performance and add implementation complexity. (Abadi et al., 2016; Hamida et al., 2023)

2.2.2 Dropout with L2 regularisation

Dropout with L2 regularisation works more indirectly. Srivastava et al. describe dropout as randomly removing units and their connections during training, while Dos Santos and Papa explain that regularisation helps prevent CNNs from overfitting. In terms of the attack, this is important because models that overfit or memorise training samples too strongly may reveal more about them through gradients. By reducing memorisation and encouraging better generalisation, dropout with L2 regularisation may weaken how closely the gradients reflect the original training image. In practice, this means the attacker may still reconstruct something, but the recovered image may be less accurate or less detailed. The drawback is that dropout with L2 regularisation is not a formal privacy defence. It is mainly designed to reduce overfitting, not to guarantee privacy. This means its effect on gradient leakage is indirect and less certain than Differential Privacy. While it is simpler and easier to implement, it cannot be claimed to stop reconstruction attacks completely. **(Srivastava et al., 2014; Hamida et al., 2023)**

2.2.3 Pixel-level face de-identification

Face de-identification methods are especially relevant to this project because the proposed defence is based on pixelation. Cao et al. show that face de-identification methods include pixel-level techniques such as masking, blurring and pixelation. In terms of the attack, this works differently from the other defences. Instead of protecting gradients mathematically or encrypting them during transmission, it changes the original image before training. This means that if leakage happens, the attacker may still reconstruct an image, but the reconstructed result should contain less sensitive facial detail. For this project, that is very important because the proposed defence aims to reduce how revealing the reconstructed image is, rather than stop all reconstruction entirely.

The drawback is that if the model is trained on more blurred or pixelated images, the training data may become less useful to the model, which can reduce overall performance or accuracy. This type of training

will only work in a type of training that is background people in a portrait or wider visual features, and also influences NPC-like game characters. **Dul (2022)**

2.3 Supporting background defence

This section discusses some of the existing defence approaches that are relevant to the literature but are not the final proposed defences in this project. They are included to raise a wider awareness of methods and to support the discussion of how privacy leakage may be reduced in different ways. In this project, these methods are treated as supporting background existing defences.

2.3.1 Data augmentation

Data augmentation works by creating altered versions of the original training images, such as rotated, cropped, flipped, or slightly modified inputs. This helps the model learn from a wider range of image variations instead of memorising one exact version of each sample. In that sense, it may reduce privacy leakage indirectly because a model that relies less on exact training examples may reveal less specific information during reconstruction attacks.

The drawback is that data augmentation is not a direct privacy defence. It is mainly designed to improve generalisation and reduce overfitting, so its effect on gradient leakage is indirect and not guaranteed. Even if augmented images are used, an attacker may still recover sensitive visual information if the gradients remain informative enough. **(Hamida et al., 2023)**

2.4 Conclusion

This chapter has shown that the main problem in this project is that the private image data may still be exposed during machine learning, even when the original image is not directly shared. The literature shows that gradient leakage is the most relevant threat for this project because if the gradients are leaked, the attacker can reconstruct an approximate version of the original training image. Model inversion supports this wider concern by showing that trained models may also reveal sensitive visual information through their outputs. Together, these studies make it clear that image privacy leakage is not just a theoretical weakness in machine learning, but a genuine privacy risk that can affect real people's data.

The chapter also showed that this issue is important outside the research literature as well. Real-world cases were included to give context to the problem. The Clearview AI case was especially relevant because it showed how real people's images could be collected at very large scale and used in facial-recognition systems without meaningful awareness or control from the individuals represented in those images. This helped connect the technical discussion of leakage attacks to a wider privacy concern around how image data is gathered, stored and used in AI systems in practice.

The main argument across the literature is that existing defences reduce the amount of useful information available to the attacker by leaked gradients. Reduce the clarity of the training image to make the attack more difficult to reconstruct. Differential Privacy is the strongest formal defence because it reduces how much sample-specific information is revealed in gradients, but its drawback is that it can reduce utility and efficiency and requires careful privacy-budget management. Dropout with L2 regularisation is easier to use and may reduce memorisation, but its drawback is that it is only an indirect defence and does not guarantee privacy. Variational Bottleneck-based methods can make gradient inversion harder by disrupting the attacker's optimisation process, but their drawback is increased complexity and extra computational cost.

The findings from this review also explain why the proposed direction of this project is reasonable. Face Pixelation During Training is supported by the face de-identification literature as a simple pixel-level method that may reduce how much sensitive facial detail is available to the model and, as a result, reduce how revealing a leaked reconstruction becomes. At the same time, the literature also makes its drawback clear. If the model is trained on more blurred or pixelated images, the data may become less useful and model performance may drop. It is also mainly suitable for specific training contexts where the face is the main sensitive feature. If other privacy-relevant information remains visible, such as background people in a portrait or wider visual features that may still influence NPC-like game characters, then face-only pixelation may have limited effect. This means the proposed defence should be understood as a lightweight way to reduce leakage in certain settings rather than as a complete solution.

Overall, the most important finding from this literature review is that privacy-preserving methods should be judged by how much they reduce the impact of leakage, not by assuming they remove the risk entirely. That is significant for this project because it directly shapes how the prototype should be designed and evaluated. The literature supports using gradient leakage as the main attack model, treating model inversion as supporting background, comparing established defences such as Differential Privacy and dropout with L2 regularisation, and including supporting background approaches such as Variational Bottleneck and data augmentation to show awareness of other possibilities. Based on these findings, the project can now move forward with a clear direction: a baseline leakage model should be compared against defended versions, and the evaluation should focus on whether reconstructed outputs become less accurate, less detailed, and less identifiable under the proposed defence. **(Rigaki and Garcia, 2023; Zhu, Liu and Han, 2019; Geiping et al., 2020)**

3 PRODUCT REVIEW

3.1 Introduction

Being a research prototype rather than a commercially released software package, this section will not review work directly related to the specific technical methodology but will provide an overview of the existing work in the field of activity recognition, a description of how each system works, its strengths and weaknesses, and how it influenced the final system developed for this research.

Two methods were chosen for further analysis: the gradient inversion method from "Inverting Gradients - How Easy Is It to Break Privacy in Federated Learning?" as it is the core idea of the attack side of the project, and the face de-identification method presented in the face privacy literature as it is closely related to the defence proposed in this project. They were chosen because they match the problem identified in the literature review that private images could be reconstructed from gradients, so any useful defence needs to reduce how much sensitive detail can still be recovered.

This section is also informed by the project proposal file, which describes the original idea of training a model on celebrity face images, reconstructing images from gradients, and comparing reconstruction quality between original-image and degraded-image training. The proposal originally pointed towards a ResNet-based setup and an existing inversion-style implementation, before the final prototype was simplified for practical reasons.

The comparison in this section is based on six simple criteria: what the approach is for, how it works, how relevant it is to the privacy problem, its strengths, its drawbacks, and how useful it was in shaping the final prototype.

3.2 Product 1 Gradient inversion reconstruction approach

3.2.1 Overview

The first approach reviewed here is the gradient inversion method presented in Inverting Gradients – How Easy Is It to Break Privacy in Federated Learning*?* by Geiping et al. This is the most important existing approach for the project because it directly addresses the main problem being studied: whether gradients can leak enough information for an attacker to reconstruct private training images.

The paper shows that federated learning does not automatically guarantee privacy just because raw data stays on the device. Instead, the gradients shared during training may still contain enough information to recover the original input. Geiping et al. show that high-resolution images can be reconstructed from gradients, even for trained deep networks, and they also show that averaging gradients across several images or iterations does not necessarily protect privacy.

This was especially important for the present project because the project proposal already followed a very similar line of thought. It described training a model on celebrity face images, extracting gradients, reconstructing images, and comparing the results between original-image and degraded-image training. It also pointed towards a ResNet-style model and an existing inversion-style implementation as the original inspiration.

3.2.2 How the attack works

This approach works by treating image reconstruction as an optimisation problem. Instead of directly accessing the original image, the attacker starts with dummy input data and updates it step by step until the dummy gradients become close to the real leaked gradients. If this optimisation works well enough, the dummy image gradually becomes similar to the original training image.

What makes this important is that it shifts the privacy problem away from raw data sharing alone. It shows that the training process itself can become a source of leakage. For this project, that was a major reason for reviewing this approach so closely. It gave the project a practical and visual way to demonstrate privacy leakage, rather than only discussing it in theory.

3.2.3 Relevance to the project

This approach is highly relevant because the project is specifically about whether **an attacker can reconstruct an image if gradients are leaked**. That makes the gradient inversion method a direct match for the problem being studied. It also fits well with the project's research-based nature, because it allows the privacy risk to be demonstrated in a controlled and understandable way.

The proposal file supports this directly. It already framed the project around training on face images, capturing gradients, reconstructing outputs, and comparing privacy leakage under different input conditions. So this reviewed approach was not just a useful background — it was one of the main foundations that helped shape the final prototype idea

3.2.4 Strength

The biggest strength of this approach is that it makes the privacy problem very clear. Instead of talking about abstract risk, it produces reconstructed images that can be inspected directly. That makes the leakage issue much easier to understand. Geiping et al. specifically argue that earlier attacks often gave a false sense of security because they only worked in simplified conditions. Their work shows that leakage can still happen in stronger and more practical deep-learning settings. A third strength is that it gives the project a strong baseline. Since the project needs an undefended case before it can test a defence, this approach gives a clear way to build that comparison.

3.3 Product 2 Face de-identification / pixel-level defence approach

3.3.1 Overview

The second approach reviewed here is the face de-identification method discussed by Cao et al. This is the most relevant defence-side comparison point for the project because the proposed defence, Face Pixelation During Training, is closely related to the pixel-level methods described in that literature.

Cao et al. divide face de-identification methods into pixel-level, representation-level, and semantic-level techniques. Pixel-level methods include masking, blurring, and pixelation, which directly alter the facial image itself before further processing. This makes the approach highly relevant to the present project, where the defence also works by reducing facial detail before training so that leaked reconstructions become less revealing.

3.3.2 How the approach works

This approach works by changing the image itself rather than only changing the learning algorithm or protecting gradients mathematically. If the face is blurred, masked, or pixelated before training, then the model receives less exact facial detail. As a result, even if leakage happens later, the reconstructed image should contain less sensitive identity information.

This was important for the project because it matches the logic of the proposed defence very closely. The aim is not necessarily to stop leakage from happening at all, but to reduce how much sensitive facial detail can still be recovered if leakage takes place.

3.3.3 Relevance to the project

This approach is highly relevant because the project does not only need to show that leakage is possible. It also needs to test whether a simple defence can reduce how revealing the reconstruction becomes. A pixel-level method such as face pixelation is especially suitable for that because the effect can be seen directly in the image.

3.3.4 Drawbacks

- The main drawback is that it may also reduce the usefulness of the data. If the face is heavily blurred or pixelated, the model may learn less effectively from the image, and that may reduce performance.
- A second drawback is that this kind of defence works best only in certain model trainings. It is most suitable when the face is the main privacy-sensitive feature. If other sensitive information is still visible, such as background people in a portrait or wider visual features that may still influence NPC-like or character-style outputs, then face-only pixelation may have a limited effect.

3.3.5 Useful for the final prototype

This review was very important in shaping the proposed defence used in the project. It showed that pixelation and related techniques already exist in the privacy literature, which helped give the defence a stronger academic basis. At the same time, it also made the limitations clearer, which helped keep the project realistic.

3.4 Key findings

Criterion	**Baseline**	**Face de-identification/ pixel-level defence**
Main purpose	Reconstruct image leaked gradients	Reduce visible identity before or during training
How it works	Optimises dummy inputs so their gradients match leaked gradients	Alter the facial image using making, blur or pixelated
Strengths	Strong demonstration of leakage, realistic threat, clear baseline	Simple, visible, practical, suitable for a proof-of-concept

Drawback	Harder to reproduce fully, more computationally demanding	May reduce data usefulness, works best in specific contexts, does not fully prevent leakage
Relevance to the project	Directly shaped the baseline leakage model	Directly shaped the proposed defence

The gradient inversion approach provided a strong baseline because it showed clearly how leakage can happen. The face de-identification approach provided a practical defence direction because it offered a simple way to reduce how revealing the reconstructed outputs might become.

3.5 Conclusion

This review showed two main things. First, gradient inversion is a strong enough existing approach to justify the baseline attack in this project, because it shows that leaked gradients may allow private training images to be reconstructed. Second, pixel-level face de-identification methods such as blur and pixelation are useful as a defence direction because they may reduce how much sensitive facial detail is still visible in the reconstructed output. Pixel-level defences are simpler and easier to test, but they may reduce the usefulness of the training data and are more suitable for certain image settings than for all cases. These points were important in shaping the project, because they supported using a smaller baseline leakage demonstration and a simpler proposed defence that could be compared clearly in a proof-of-concept setting.

4 REQUIREMENTS ANALYSIS AND DESIGN

4.1 Introduction

The prototype in this project was built to demonstrate one main idea: that leaked gradients may allow the attacker to reconstruct an approximate image, and then the proposed defence may reduce how revealing that reconstruction becomes. In order to compare both, the system was designed around two pipelines: a baseline leakage pipeline and the defended pipeline. To make that comparison possible, the prototype needed clear requirements for dataset preparation, model training, gradient capture, reconstruction, and output comparison.

The main idea for this prototype came from the literature review, especially the work on gradient inversion that showed private images could be reconstructed from the leaked gradients. The literature "Inverting Gradients – How Easy Is It to Break Privacy in Federated Learning" includes stronger architectures such as ResNet, but in this project, a CNN was used instead. This decision was made to keep the implementation simple and more manageable for a proof of concept, while still preserving the core aim of demonstrating gradient leakage and comparing with the proposed defence with the baseline case

4.2 Requirements Analysis

4.2.1 Requirements

The requirements for the prototype were mainly taken from the literature review. The literature review was the starting point because it showed that gradients can contain enough information for the attacker to reconstruct private training images. That means the prototype ned to do more than train a model. It needed to demonstrate the leakage process by training on an image, extracting gradients and attempting reconstruction from those gradients.

The literature also shows that defences are usually discussed as ways of reducing leakage rather than fully preventing it. Because of that, the prototype needed a second condition as well as the baseline. The defended version is needed to show the effect of the proposed defence and compare it between the baseline and the defended version.

The proposal originally pointed towards a ResNet-based setup and an existing inversion-style approach, but instead picked a simple prototype using a CNN. The original project proposal supported this direction. It described a comparative setup in which a model would be trained on face images, gradients would be captured, images would be reconstructed, and the process would then be repeated on degraded versions of the same input. Even so, the proposal was still useful because it defines the key experimental

stages clearly, which are preprocessing, model training, gradient extraction, reconstruction and comparison between the baseline and the proposed defence pipeline.

The product review helped refine these ideas further. The gradient inversion approach reviewed earlier confirmed that a reconstruction-based baseline was the right way to demonstrate the privacy problem. The face de-identification review then supported the use of a simple image-level defence. This was important because the model needed to produce a clear visual comparison.

This prototype only needed the features required to support the research questions. It did not need a full federated learning platform. What was needed was a clear baseline attack and a defended version of the same process, and a way to compare the outputs both visually.

4.2.2 Assumptions and Constraints (if applicable)

One important assumption was that the prototype would be used as a **proof-of-concept** rather than a full deployable system. The aim of the project was to support the research by demonstrating the leakage problem clearly and then comparing it with the proposed defence. Because of that, the system only needed the features required for that comparison. It did not need a full user interface, a complete federated learning environment, or commercial deployment features.

Another assumption was in model choice. The main idea for the prototype came from the literature review, especially Inverting Gradients – How Easy Is It to Break Privacy in Federated Learning?, where stronger architectures such as ResNet were used to study gradient inversion in more realistic settings. However, the final implementation in this project used a CNN instead. This was a practical constraint rather than a theoretical one. A CNN was chosen because it was simpler to build and easier to train. It shows that gradients may leak image information and allowing a direct comparison between the baseline and defended case.

This project was also constrained by time and computational resources. Gradient inversion can be expensive when the prototype is a larger model or with more images, or longer optimisation steps are used.

This meant the prototype had to stay focused on a small, controlled experiment rather than a large-scale system. That same constraint also affected the number of defences that could be implemented. Although the literature includes methods such as Differential Privacy and variational bottlenecks, it was not realistic to implement all of them into the prototype. Because of that, the main implementation focused on the proposed image-level defence, while the other methods were treated as existing literature-based defences.

The project was designed around a comparison between two pipelines only: a baseline leakage case and a defended case. That decision helped keep the work clear and focused, and it also means that the prototype is limited to demonstrating one specific attack scenario rather than covering every possible privacy AI image system.

This assumption and constraints were more manageable, kept the implementation aligned with the research aim, and made it easier to focus on the main comparison the project was trying to demonstrate.

4.2.3 Functional Requirements

The functional requirements were identified from the actual prototype workflow, then prioritising using the MoSCow method. In this project, the prototype needed to do enough to demonstrate the leakage problem clearly and compare both results in a controlled way. The main functions were dataset preparation, image processing, CNN training, gradient extraction, dummy image initialisation, gradient inversion, visual comparison, and Metric-based evaluation.

4.2.4 Must have

- **Load image data set –** the prototype needed a real image input for the both baseline and the defended experiments.
- **Preprocess images-** resizing and tensor conversion is necessary so the CNN could use the images in a fixed format.
- **Train CNN model –** Gradient could only be generated through a real training process.
- **Extract gradient from the model –** this was important because main source of the project is gradient leakage.
- **Initialise dummy data –** this is required to reconstruct an approximate image from the leaked gradients.
- **Proposed defence –** this is with the same reconstruction process were the defended case could be compared fairly with the baseline.
- **Compare baseline and defended pipeline –** this was essential because the project is based on testing whether the defence reduce leakage.

4.2.5 Should have

- **Displayed original and reconstructed image –** this is image is placed sideby side inorder to display the leakage effect visually.

- **Calculate MSE, PSNR and SSIM –** these metrics supported the visual comparison with numerical evidence.

4.2.6 Won't have

- A full federated learning environment
- A complete user interface
- Production level infrastructure

The must-have requirements were the functions directly needed to demonstrate gradient leakage and compare the proposed defence with the baseline. The should-have requirements improved the clarity and quality of the results, but the core experiment could still run without all of them.

4.2.7 Non-functional Requirements

The non-functional requirements describe the qualities the prototype needed to have so that the experiment could be carried out clearly within the limits of the project. These were prioritised using the MoSCoW method

4.2.8 Must-have

- **Clarity and understandability-** the prototype needed to be simple enough to explain clearly in the report.
- **Computational manageability –** the experiment needed to run within the available time and hardware limits.
- **Readable outputs – the** reconstructed and comparison image results needed to be easy to interpret
- **Consistent comparison –** the baseline and defended pipelines needed to follow the same overall process, so the comparison remains fair

4.2.9 Should have

- **Reasonable reproducibility** – repeating the experiment under the same settings should produce comparable results.
- **Module structure** – the baseline and the defended version needed to be easy to describe them separately.
- **Usefulness of defended input –** the defended version should keep enough image information to make the process meaningful.

4.2.10 Won't have

- **Real-time performance**
- **Commercial infrastructure**

4.3 Design

The final of design of the prototype was kept simple so that the main privacy leakage problem can be demonstrated clearly. Instead of building a larger system, the design focuses on one controlled comparison between a baseline leakage case and a defended case. This was the most suitable design for the project because the main aim was to show that gradients may leak image information, and then test whether the proposed defence could reduce how revealing the reconstructed image becomes.

4.3.1 Model design overview

The prototype was designed with two pipelines. The first was the baseline leakage pipeline. In this version, the original image was used as the model input, the model passes the image through a training process, and then the model produces the gradients, and reconstruction was attempted from those gradients. The second pipeline follows the same process, applying the proposed defence image to the model as input, and it made it possible to compare the reconstruction from the original image with the reconstruction from the defended image.

This design was chosen because it gives the project a clear comparison between the two pipelines; the system did not need those extra features unrelated to the research question. It only needed enough design to support training, gradient extraction, reconstruction, defence application, and output comparison.

4.3.2 System Architecture

At a high level, the system follows this structure

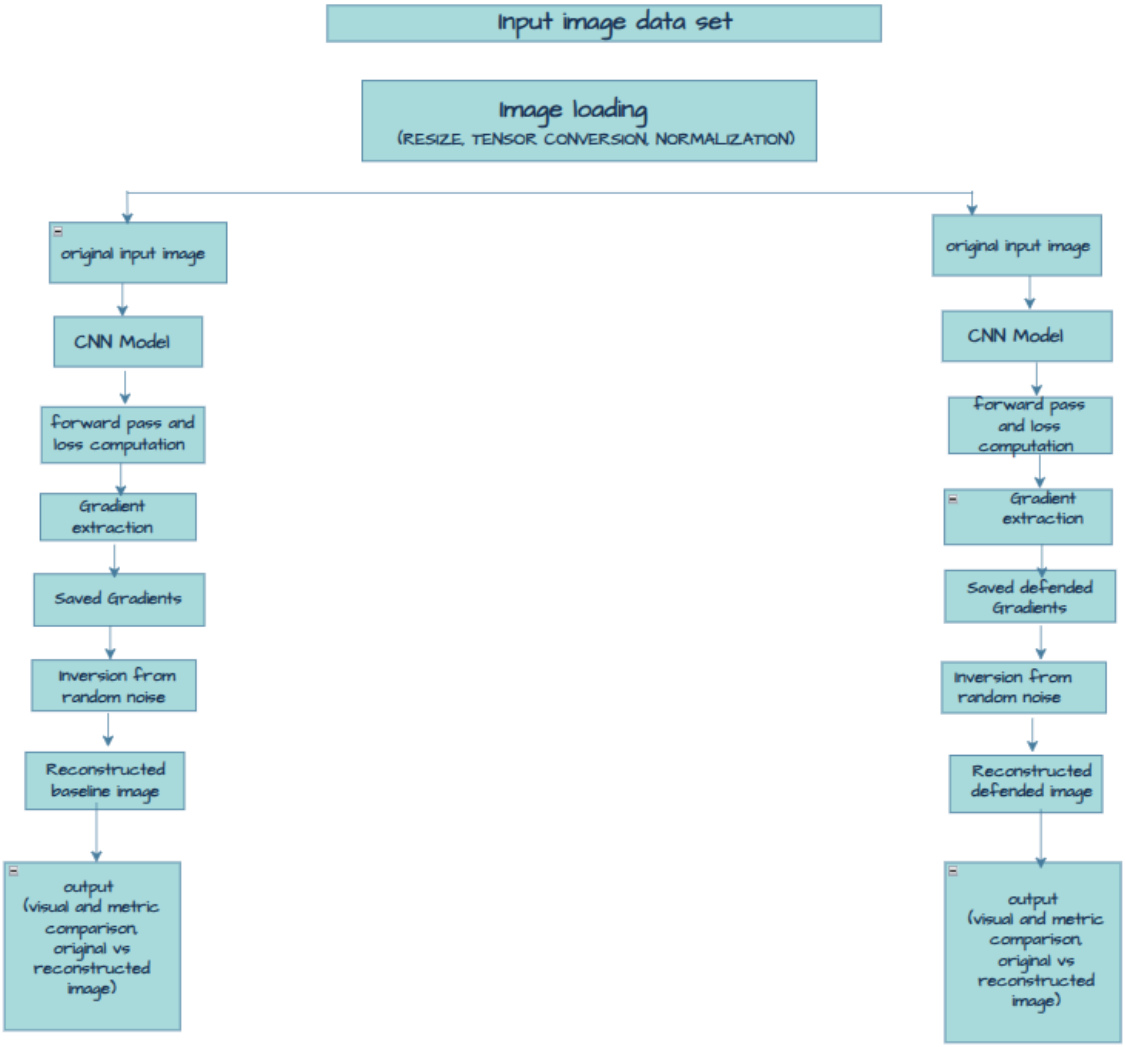


Fig 1.2 System architecture

The architecture was useful because both pipeline followed the same general process. And that made the comparison fair and the defence is the main intended difference between these two conditions

FLOW DIAGRAM

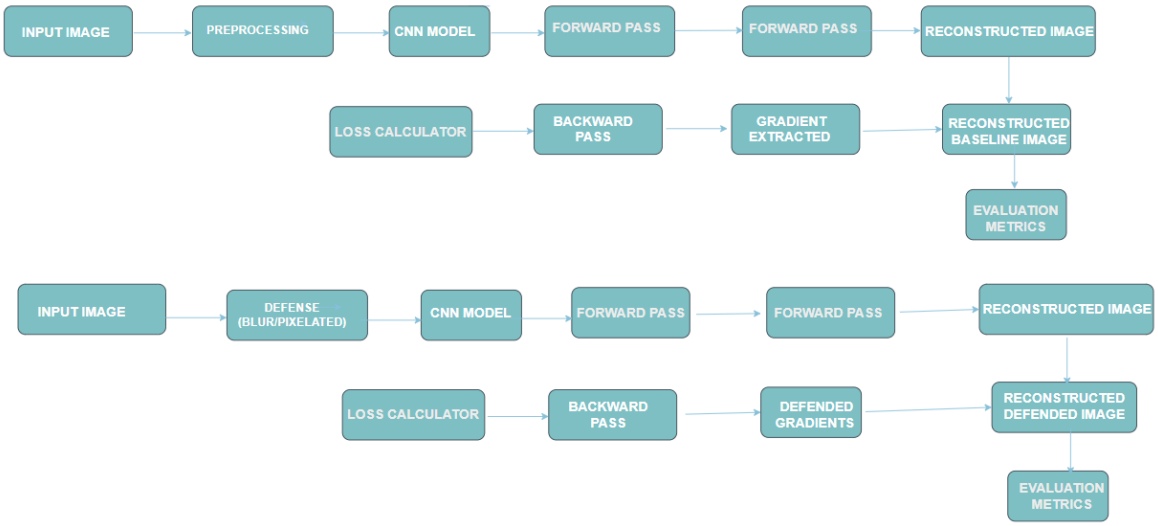


Fig 1.3 Flow diagram

4.3.3 Data set design

The data set was kept simple, images were loaded from a local folder, resized into a fixed size, and then converted into a tensor format and normalised before being passed into the model. This was necessary because the CNN required a consistent input format. It also made the experiment easier to repeat and compare.

4.3.4 Model Attack design

The final implementation used a **CNN** model. The main idea came from the literature on gradient inversion, especially studies using stronger architectures such as ResNet. However, the final design used a CNN instead because it was simpler and easier to manage within the project scope.

The attack design followed by the gradient inversion process used in the literature. The model processed the image, and gradients were extracted and saved. Reconstruction then started from a random dummy image. That dummy image was updated repeatedly so that its gradients became closer to the saved gradients. Over time, this produced an approximate reconstruction of the original input.

This design was important because it matches the main privacy problem in the project. The goal was not just to train a model but to show that even if the attacker gets the gradients, they can reconstruct image information.

4.3.5 Defence design

The proposed defence was designed as an image-level step applied before gradient extraction. Instead of changing the Gradients directly, the defence changed the image first so that the model trained on less detailed version of the face. The same general reconstruction process was then repeated. his design choice was made because it was simple, visual, and easy to compare with the baseline. It also matched the face de-identification literature, which shows that pixel-level methods such as blur and pixelation can reduce visible facial detail before further processing. In this project the defence is not expected to stop the leakage completely but its purpose was to reduce the how much sensitive data information remained in the reconstructed result.

4.4 Conclusions

This chapter identified the main requirements of the design choice and the prototype. The most important requirement was the ability to load and process the image data, train a CNN model, Extract gradients, reconstruct an approximate image from those gradients, and also apply the proposed defence, and compare the baseline with the defended results. Non-functional requirements such as clarity, computational manageability, and a fair comparison. A CNN was chosen instead of a more complex ResNet-based setup to keep the implementation simpler and more manageable. The proposed defence was applied at the image level before gradient extraction, which made it easier to test and compare visually.

Several assumptions and constraints shaped the chapter as well. The prototype was treated as a proof-of-concept, gradients were assumed to be available to the attacker, and the project scope was limited by time, computational cost, and the need to keep the system focused.

Overall, the requirements and design decision kept the prototype closely aligned with the project aim: to demonstrate clear gradient leakage and test whether the proposed defence could reduce how revealing the reconstructed output became. This provided the foundation for the next chapter.

5 IMPLEMENTATION

5.1 Introduction

The prototype implemented in this project was developed as a proof of concept to demonstrate the gradient leakage and compare the proposed defence against the baseline case. The final system was built around two pipelines; the first uses the original image to generate gradients and attempt reconstruction from them. The second pipeline also uses the same process after applying the proposed defence to the image before gradient extraction. This made it possible to compare how much visual information is being recovered in both conditions. The main idea was influenced by the gradient inversion literature reviewed earlier, although the final implementation used a simpler CNN-based setup rather than a larger ResNet-based model to keep the prototype manageable within the scope of the project.

5.2 Code outline

The source code for the prototype was organised around the main stages of the experiment. The implementation was developed mainly from scratch, but it was clearly influenced by the gradient inversion literature and by the general workflow described in the original project proposal. And the code adapts the core idea into a smaller proof of concept suited to the aims of the project.

- Data setup – loading the chosen image into the simple dataset structure
- Pre-processing – resizing the image, converting in to tensors, and preparing it for model input.
- Model definition – Creating the CNN used in the experiment.
- Training and gradient capture – forward pass, compute the loss, and then run back and save the gradients
- Gradient inversion – reconstructing an approximate image from the leaked gradients using optimisation from the random noise.
- Proposed defence: repeating the same process after applying the image pixelation before gradient extraction
- Evaluation and comparison of display output – comparing the original image and the reconstructed images visually and numerically using MSE, PSNR, and SSIM

5.3 Baseline pipeline implementation

The baseline pipeline was implemented to demonstrate the privacy leakage problem in its simplest form.

The process began by loading the image dataset from a local folder and preparing it for use in the model. The implementation notes show that the dataset was unzipped, moved into a folder structure, and then loaded through a custom single-class dataset. And then the image is resized into a fixed size and then converted into a tensor format before being passed into the CNN.
Once the model had been defined, the training setup was created with a loss function and an optimiser. A single image sample was loaded into the system, passed through the model, and then the loss value was computed, and the gradient was generated and saved. This stage was important because the project is specifically concerned with what happens when gradients are exposed. The implementation therefore, treated the saved gradients as the information available to the attacker.

The reconstruction stage followed the same general gradient inversion idea found in the literature, where a dummy image was initialised from random noise and then updated iteratively. At each step, the gradients produced from the dummy image were compared with the saved gradients from the real image, gradually making the dummy image resemble the original input more closely. This will produce an approximate reconstruction based only on gradients rather than direct access to the original image. The final output from the baseline pipeline consisted of a visual side-by-side comparison between the original image and the reconstructed image, along with numerical measures such as MSE, PSNR, and SSIM.

5.4 Defended pipeline implementation

The defended pipeline followed the same overall structure as the baseline pipeline, but with one difference: the image was degraded or pixelated before training or before gradients were created. The defended pipeline produced the same type of output as the baseline pipeline: a reconstructed image and numerical comparison scores. In this way, the implementation could show whether the defence reduced how much useful visual detail remained in the leaked reconstruction.

5.5 Testing

To ensure correct operation of the prototype and that reasonable output was obtained from the baseline and defended models used for comparison, a number of tests were conducted. Since this was just a research proof-of-concept and not a large software application, testing was limited to the checking of the main workflow (preparation of dataset, pre-processing, CNN execution, Gradient extraction, Reconstruction, defence application and comparison of output).

The three levels of testing were unit testing, integration testing, and output testing. Unit tests were useful for checking that the smaller components of the program worked as required, integration tests were useful for checking that components worked as required when executed together and in the proper order and output tests were useful because the main requirement of the prototype was to display the

reconstructed images and compare the two pipelines. The tests were performed in Google Colab with the same Python and PyTorch environment that was used to create the prototype.

5.5.1 Unit Testing

Unit tests were defined for each element of the prototype. At the dataset loading phase, the resizing, conversion of tensors and normalization of the data were each subjected to a unit test to ensure the model would receive valid data.

The next step was to test the CNN. It should accept the processed image as input, create an output without shape issues, support loss calculations, and eventually the extraction of gradients. The last one is very important because the entire privacy leakage in this experiment relies on the existence of gradients available to the reconstruction mechanism.
The reconstruction part was validated by checking whether the dummy image was correctly initialized, whether the loop was correctly executed in Colab , and whether the dummy image changes at every iteration. Furthermore, the metric calculation part was validated by checking whether the MSE, PSNR and SSIM can be calculated from the reconstructed outputs.

5.5.2 Integration Testing

Integration tests checked whether the main components of the prototype could work together in the correct order. For example in Google Colab this checked whether the image loading and pre-processing to the CNN and the gradient extraction, image reconstruction and the image output display did not break the program's workflow. The complete pipeline was tested for both base and defended runs, because this project was highly reliant on a side by side comparison, and it was necessary for each of the steps to work.

5.5.3 Usability Testing

Usability testing was kept simple because the prototype was never designed to be a user-facing system; the main question was merely whether or not the outputs made sense to the user. In practice, this would be the ability to show the original and reconstructed images side by side and the clarity with which numbers for the MSE, PSNR, SSIM and so on could be displayed for visual inspection, as the value of the prototype depends on how readily the leakage result and the defended result could be visualized and analysed.

5.6 Conclusion

At the conclusion of the implementation and test run, the final result confirmed the prototype had worked as a small proof of concept that the baseline was able to derive the model's gradients and reconstruct an input image from the model, and that the defended version could do the same after implementing the proposed defence strategy. The main stage of the system was

confirmed as working together, with output that was clear enough to follow-up on. The prototype had its limitations as well. It used a simplified version of the model with a CNN architecture, and was limited in that it was tested in the smaller google Collab environment. Still, we believe the prototype was sufficient to achieve the main aim of the project as it was in its capacity to show the gradient leakage and compare the result of defended images with the baseline. In the next chapter we move on to the evaluation of the prototype outputs.

6 RESULT AND EVALUATION

6.1 Introduction

This chapter shows and compares the results produced by the prototype and evaluates the results in relation to the project's aim, objectives, and requirements. The prototype was designed only to compare the two conditions, that is, a baseline leakage and the defended case. In the base pipeline, gradients are extracted from the original image by passing through the CNN and reconstructing an approximate version of the image. In the defended version, the same process was repeated after applying the defence strategy as in input to the model in order to extract the gradient.

The result section reports the raw outputs, defended output, and compares them with numerical measures such as MSE, PSNR, and SSIM, and the evaluation section considers how well the result addresses the project objectives, main strengths, and limitations.

6.2 Results

The reconstruction results may vary slightly between runs because the gradient inversion process depends on iterative optimisation, where matches between the saved gradients and the dummy gradients. Images can slightly differ each time the experiment runs. As a result, the matric values of the images may not be identical in every run.

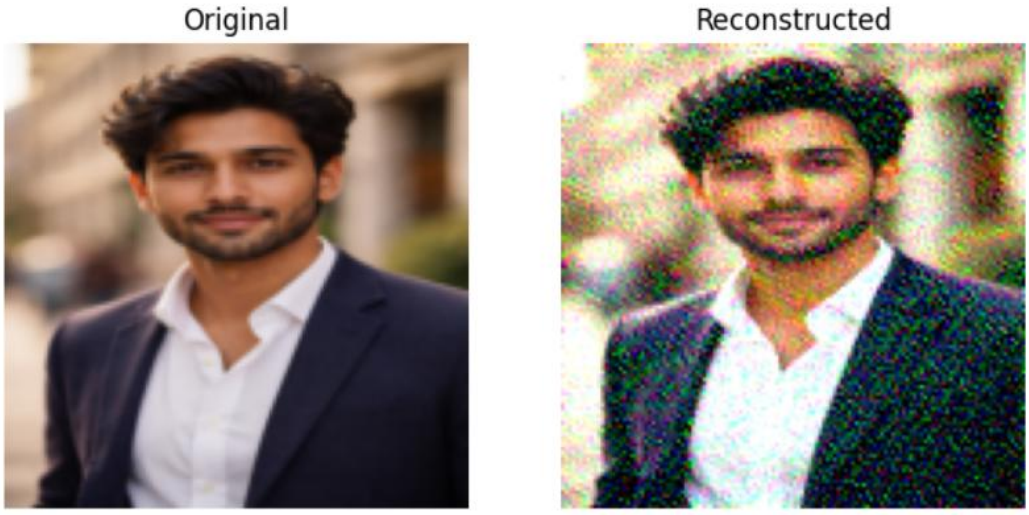


Fig 2.1 baseline reconstructed result

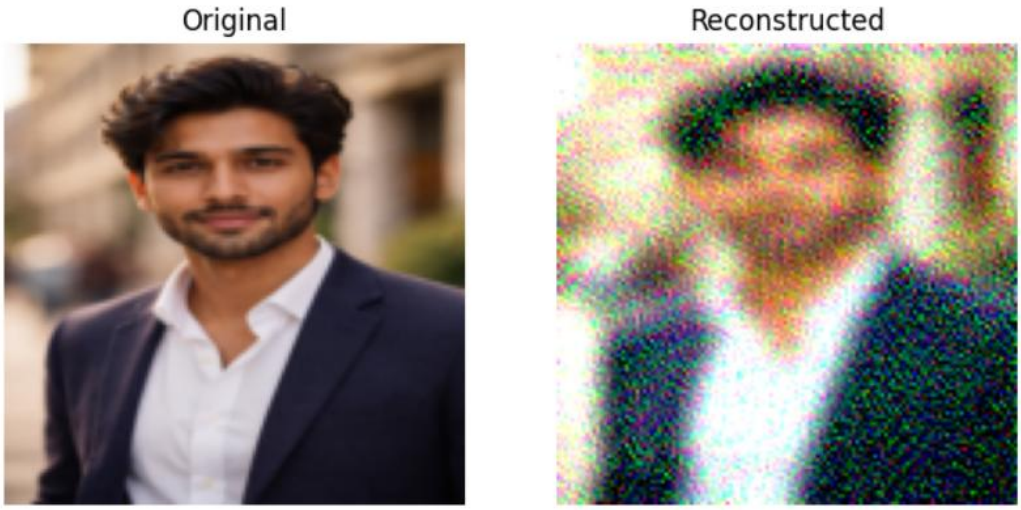


Fig 2.2 Defended reconstructed result

6.2.1 Baseline reconstruction Results

The baseline pipeline produced a reconstructed image from the leaked gradients using the original input image. The output of the baseline shows that the reconstruction from the leaked gradients was possible, and it can retain visual information from the original image. The reconstructed result was not that accurate, but it still retained a recognisable overall structure of the image, like general face shape, hairstyle, and clothing outline, and parts of the background were also visible, although the image also contained noticeable noise.

The numerical result

Metric	Baseline result
MSE	0.042849
PSNR	13.68 db
SSIM	0.4148



Fig 2.3 baseline reconstructed result with metrics

The output shows that the reconstructed image still matches the original image, and it is enough to demonstrate that the useful information had been leaked through the gradients.

6.2.2 Defended reconstruction results

The defended pipeline repeats the same process after applying the proposed defence strategy before gradient extraction. The reconstructed output from this pipeline appeared more blurred and less detailed than the baseline reconstruction. Outline of the person and colour information were still visible.

Visually, the defended output shows a weaker reconstruction than the baseline case. This made the defended image less revealing, which is the main effect of the proposed defence that was intended to achieve. In other words, reconstruction was still possible, but the result appeared less useful.

Metric	Defended result
MSE	0.059932
PSNR	12.22 db
SSIM	0.2569



Fig 2.4 Defended reconstructed result with metrics

6.2.3 Comparison of the output

A direct comparison between the two outputs shows a clear difference in visual quality. The baseline reconstruction leaked more facial detail and general image structure, while the defended reconstruction was more blurred and less identifiable.

Metric	Baseline result	Defended result
MSE	0.042849	0.059932
PSNR	13.68 db	12.22 db
SSIM	0.4148	0.2569

The comparison is very important because the project was designed as a two-condition experiment rather than a single reconstruction demonstration. The baseline result showed that applying the proposed defence changed the final reconstruction and reduced its clarity.

6.2.4 Results Conclusion

The result shows that the prototype successfully produced image reconstructions from leaked gradients in both the baseline and the defended pipeline. The most important finding from the results section is that the baseline pipeline demonstrated visible leakage of image information, while the defended pipeline has created less detailed image reconstruction. These outputs give the basis for the next section, where the results are evaluated more critically.

6.3 Evaluation

The prototype was evaluated using both visual inspection and numerical comparison. This was appropriate because the project is concerned with image privacy leakage, so the quality of the reconstructed image is important.

These are the three measures that were used:

- **MSE –** which measures pixel-level difference between the original and reconstructed image
- **PSNR –** which measures reconstruction quality, where the higher values indicate better similarity.
- **SSIM –** which measures structural similarity and is particularly useful for judging image quality.

These metrics were suitable because the project not only asks whether a reconstruction can be produced, but also how much useful information was leaked while reconstructing the leaked gradients.

6.3.1 Evaluation against project objectives

The first objective of the project was to demonstrate that the gradients may leak enough information for an image to be reconstructed. The baseline output showed that the reconstructed image preserved enough structure, colour distribution, and general facial information to the attackers, which confirms that leakage had occurred. Even though the reconstruction was noisy, it was clearly more meaningful than random noise.

The second objective was to test whether the proposed defence reduces the impact of the leakage. The defended reconstruction appeared more blurred and less visible details than the baseline reconstruction, which suggests that the defence reduces how much useful information could be recovered. This matches the intended role of the defence in the project. The aim was not to prevent the attack but to reduce the impact of the ability to reconstruct the image from the leaked gradients.

The third objective was to compare the baseline and the defended pipeline clearly. This was achieved through both visual outputs and metric comparison, and the side-by-side result made it possible to differentiate directly.

6.3.2 Strength of result

One strength of this project is that it produced both visual and numerical evidence. The prototype not only claimed that leakage was possible. The reconstructed image provided measurable outputs such as MSE, PSNR, and SSIM. This made the finding more convincing and easier to evaluate.

Another strength is that the project used a direct comparison between the baseline and the defended pipelines. This made it possible to assess the proposed defence in a meaningful way. Because the process stayed the same for both of the cases.

A further strength of the prototype is that the model remained close to the project's aim. It did not try to become a large system or an unrelated software product. Instead, it stayed focused on the central privacy issue and produced results that were directly relevant to that question.

6.3.3 Limitations and Unexpected Findings

The result also has clear limitations. The reconstructed quality was limited, especially in finer detail. This means the prototype demonstrated leakage mainly in a general way rather than full high-accuracy facial recovery.
Another limitation is model simplicity. The final prototype used CNN rather than a stronger architecture like ResNet. This was to make the model more manageable, but it also means the experiment does not directly match the complexity of the larger gradient inversion studies from the literature review.

The model was tested in Google Colab as a proof of concept. This was suitable for an academic demonstration, which means the result should not be compared with the federated learning or commercial deployment environment.

Another limitation is that the proposed defence does not prevent the leakage it only reduces the impact of retaining useful information from the leaked gradients. This was not unexpected since the literature already suggested that most of the defences are to reduce the impact rather than eliminate the full risk.

6.3.4 Comparison with the existing work

When compared with the literature, the results are broadly consistent with what earlier studies found. The main research papers showed that the gradients can reveal the private training data, and the quality of the reconstruction quality depends on the architecture and optimization process, as well as the

experimental setting. The current project supports the general conclusion, although at a smaller scale and with a simpler model, while the research papers are in federated learning and use a higher architecture like ResNet.

Existing work suggests that the privacy-preserving methods often reduce the quality of leaked information rather than stopping leakage entirely. The result from the project fit that pattern since the defended output was visibly less detailed but still not completely private.

6.4 Conclusion

The chapter shows that the prototype was able to demonstrate gradient leakage in a controlled way and also produce measurable outputs for the baseline and the proposed defence pipeline. The baseline result shows that the useful details of the image could be reconstructed from leaked gradients, while the defended result shows a weaker and less detailed reconstruction. This means the project met its main objective of demonstrating the privacy risk and testing whether the proposed defence could reduce in the impact of that risk.

At the same time, the evaluation highlighted the limits of the work. The created prototype was a small and simplified CNN model, and also produced imperfect reconstructions. Even so, the results were strong enough to support the main conclusion of the project: the gradients can leak useful image information, and simple defensive measures may reduce how revealing that leakage becomes, even if they do not remove it completely

These findings provide the basis of the final chapter, which summarises the overall project, reflects on the contribution and the limitations, and suggests future work.

7 LEGAL, SOCIAL, ETHICAL AND PROFESSIONAL ISSUES

The project has resulted in the exploration of legal, social, ethical, and professional challenges with the use of image data, leaking of private information, and reconstructing information from model gradients. Thus, this prototype, although assembled for research purposes only, presents the need for data protection when using artificial intelligence.

This was a fully synthetic portrait, created from an openly-licensed AI-generated input image, but in principle, a similar application trained on real people could have used data about their faces, which would engage the Data Protection Act 2018 and the UK GDPR. In this instance, the absence of real data in the training set avoided any potential Data Protection issues.

There is also the ethical/social issue that while this has been done for defence and research purposes, the same technology can be abused. Privacy leakage threatens the trustworthiness of AI systems by reducing confidence in how pictures are used or reused. This leads to people refraining from sharing pictures when they would otherwise do so in a trustworthy environment.

From the computing professional's perspective, this project is also associated with the BCS's code of conduct, especially its responsibility to act in the public interest and to respect privacy, security and the rights of others, which means that this work should be presented honestly by its limits and risks. This reveals that the privacy leakage is not only a technical problem but also a legal, ethical, social and professional one.

8 CONCLUSION

8.1 Summary of the work

The objective of this project is to study the privacy implications of image generative AI models via gradient leakage and reconstruction methods. The project consists of a literature survey that covers the major publications in privacy attacks and defence methodologies and the design and implementation of a small proof of concept. The prototype extracted gradients for both the baseline leakage pipeline and the defended pipeline, allowing for the reconstruction of a close representation of the image for the baseline case. In the defended case, the same process was then followed once the defendant had raised that proposed defence, for comparison purposes.

8.2 Overall outcomes

In the end, the project met all of its goals: it demonstrated that an image's gradients could reveal information, and it also constructed a prototype to show the threat. It has shown that the proposed defence indeed reduced the amount of information the output of the reconstructions contained over the baseline case, thus supporting our main hypothesis of the project that privacy leakage in AI image systems is a real threat and that simple defensive measures can reduce this threat even when they cannot completely eliminate it.

8.3 Limitations and reflection

There were also limitations: the model was a small, constrained prototype, trained with a CNN architecture and not with other architectures such as ResNet, the work was a proof of concept and not an actual federated learning system, and the defence was tested under specific conditions with image data, meaning the results are not generalizable to other AI image systems.

Further, the research was only tested as a proof of concept, not in a fully federated learning system, and the defence was only tested on images, not on other types of AI image systems.

8.4 Future Work

Future work includes running the studies with larger models and with more datasets, but also testing other privacy methods from the literature, including Differential Privacy or Variational Bottleneck methods. Another extension would be to run such studies in a more realistic federated learning environment to study leakage in more realistic settings. Another avenue you might pursue is to try

deploying the proposed defence on other types of image tasks to get a better sense of where it succeeds and where it struggles.

Overall, this project showed that privacy leakage from gradients is an important issue in AI, and improved both the understanding of the risks, and the practical challenges of reducing the risk

REFERENCES

- Xu, B., Wen, Q., Cheng, L., Hu, X., Li, T. and Sun, Y. (2025). Gradient Inversion Attack via Image-Correction-Penalty Based Over-Parameterized Regression Network in Federated Learning. *IEEE Internet of Things Journal*, pp.1–1. doi: https://doi.org/10.1109/jiot.2025.3572806.

- Fredrikson, M., Jha, S. and Ristenpart, T. (2015). Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures. *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security - CCS '15*. [online] doi: https://doi.org/10.1145/2810103.2813677.

- Gat, N. and Sharif, M. (2024). Harmful Bias: A General Label-Leakage Attack on Federated Learning from Bias Gradients. *Proceedings of the 2024 Workshop on Artificial Intelligence and Security*, pp.31–41. doi: https://doi.org/10.1145/3689932.3694768.

- Jonas, G., Hartmut, B., Hannah, D. and Michael, M. (2020). Inverting Gradients - How easy is it to break privacy in federated learning? *Advances in Neural Information Processing Systems*, [online] 33. Available at: https://proceedings.neurips.cc/paper/2020/hash/c4ede56bbd98819ae6112b20ac6bf145-Abstract.html.

- Zhu, L., Liu, Z. and Han, S. (2019). *Deep Leakage from Gradients*. [online] Neural Information Processing Systems. Available at: https://proceedings.neurips.cc/paper/2019/hash/60a6c4002cc7b29142def8871531281a-Abstract.html.

- Hamida, S.B., Hichem Mrabet, Faten Chaieb and Abderrazak Jemai (2023). Assessment of data augmentation, dropout with L2 Regularization and differential privacy against membership inference attacks. *Multimedia Tools and Applications*, 83(15), pp.44455–44484. doi: https://doi.org/10.1007/s11042-023-17394-3.

- Rigaki, M. and Garcia, S. (2023). A Survey of Privacy Attacks in Machine Learning. *ACM Computing Surveys*, [online] 56(4). doi: https://doi.org/10.1145/3624010.

- Pan, Y., Chao, Z., He, W., Jing, Y., Hongjia, L. and Liming, W. (2024). FedSHE: privacy preserving and efficient federated learning with adaptive segmented CKKS homomorphic encryption. *Cybersecurity*, 7(1). doi: https://doi.org/10.1186/s42400-024-00232-w

- Fang, L., Wang, L. and Li, H. (2024). Iterative and mixed-spaces image gradient inversion attack in federated learning. *Cybersecurity*, 7(1). doi: https://doi.org/10.1186/s42400-024-00227-7.

-
- Scheliga, D., Mäder, P. and Seeland, M. (2025). Privacy preserving federated learning with convolutional variational bottlenecks. *Cybersecurity*, 8(1). doi: https://doi.org/10.1186/s42400-024-00295-9.

- Yang, Y., Ma, Z., Xiao, B., Liu, Y., Li, T. and Zhang, J. (2023). Reveal Your Images: Gradient Leakage Attack Against Unbiased Sampling-Based Secure Aggregation. *IEEE Transactions on Knowledge and Data Engineering*, 35(12), pp.12958–12971. doi: https://doi.org/10.1109/tkde.2023.3271432.

- Zhang, R., Guo, S., Wang, J., Xie, X. and Tao, D. (2022) 'A survey on gradient inversion: Attacks, defenses and future directions', in *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)*, pp. 5678–5685 https://www.ijcai.org/proceedings/2022/790

- Yang, W., Wang, S., Wu, D., Cai, T., Zhu, Y., Wei, S., Zhang, Y., Yang, X., Tang, Z. and Li, Y. (2025). Deep learning model inversion attacks and defenses: a comprehensive survey. *Artificial Intelligence Review*, 58(8). doi: https://doi.org/10.1007/s10462-025-11248-0.

- Zhang, Y., Jia, R., Pei, H., Wang, W., Li, B. and Song, D. (2020). The Secret Revealer: Generative Model-Inversion Attacks Against Deep Neural Networks. *Thecvf.com*, [online] pp.253–261. Available at: https://openaccess.thecvf.com/content_CVPR_2020/html/Zhang_The_Secret_Revealer_Generative_Model-Inversion_Attacks_Against_Deep_Neural_Networks_CVPR_2020_paper.html.

- Xiong, P., Liu, F., Zhang, J., Lan, L., Ye, J., Liu, T. and Han, B. (2022). Bilateral Dependency Optimization. *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. Doi: https://doi.org/10.1145/3534678.3539376.

- Zhou, S., Ye, D., Zhu, T. and Zhou, W. (2024). *Defending Against Neural Network Model Inversion Attacks via Data Poisoning*. [online] arXiv.org. Available at: https://arxiv.org/abs/2412.07575.

- Dul, C. (2022). Facial Recognition Technology vs Privacy: The Case of Clearview AI. *qmro.qmul.ac.uk*. [online] Available at: https://qmro.qmul.ac.uk/xmlui/handle/123456789/80559.

- Gal Nagli (2025). *Wiz Research Uncovers Exposed DeepSeek Database Leaking Sensitive Information, Including Chat History*. [online] wiz.io. Available at: https://www.wiz.io/blog/wiz-research-uncovers-exposed-deepseek-database-leak.

- Mansoor, S. (2026). *Claude's code: Anthropic leaks source code for AI software engineering tool*. [online] the Guardian. Available at: https://www.theguardian.com/technology/2026/apr/01/anthropic-claudes-code-leaks-ai.

- Luca Beurer-Kellner, Aleksei Kudrinskii, Milanta, M., Nielsen, K.B., Sarkar, H. and Tal, L. (2026). *280+ Leaky Skills: How OpenClaw & ClawHub Are Exposing API Keys and PII*. [online] Snyk. Available at: https://snyk.io/blog/openclaw-skills-credential-leaks-research/

APPENDIX A - Project Proposal

a.1 Revised Proposal/Project Aim

The project proposal was also focused on privacy leakage in Image-based AI platforms from gradients. The main research remained the same throughout the project.
However, the early version of the project was only theoretical and also discussed several defence methods from the literature without including any practical model. The project would only explain that certain defences might work in the future, rather than showing the effects directly.

To strengthen the project, a small poof of concept was included, which made it possible to move beyond theory and demonstrate the privacy problem more clearly through the baseline and the defended pipeline. The topic remains the same in the final work, but it also implemented a defence strategy and demonstrated that in the model.

a.2 Revised Project Plan

The revised proposal kept the same focus on privacy leakage in image-based AI systems, especially gradient leakage and model inversion attacks. However, the final version became more focused and more practical than the original plan. In the early stage, many defence methods were being discussed only in theory, which meant the project would mainly explain that certain defences might work without showing them in practice. To make the project stronger, a small proof-of-concept model was added. This made it possible to demonstrate the privacy problem more clearly through a baseline leakage pipeline and a defended pipeline. The final version also narrowed its practical focus by using a CNN instead of a larger model such as ResNet, choosing Google Colab as the development environment, and refining the proposed defence into Face Pixelation During Training so that the comparison could be clearer and easier to evaluate.

a.3 Original Proposal

1.1 Aim

The aim of this project is to study and analyse how privacy leakage occurs in AI-based image generative models like ChatGPT, Gemini, focusing on two critical vulnerabilities: Gradient leakage and Model inversion Attacks
This research investigates how attackers can reconstruct user images from model gradient outputs and evaluates the effectiveness of existing and emerging defensive methods. This project provides a comparative academic analysis of these attacks and defences to identify which approaches offer the best balance between privacy protection and model accuracy.

1.2 Objectives

These projects investigate how image privacy leakage occurs in AI platforms such as Google Gemini. ChatGPT, and other platforms, analysing how attackers exploit models through gradient leakage and model inversion attacks

It also evaluates the performance of existing defence mechanisms designed to mitigate these risks. Through an in-depth literature review, both classical and modern defences- **including shadow defence, bilateral dependency optimisation (BIDO), Differential privacy (DP), Homomorphic encryption (HE), convolutional variational bottlenecks (CVB), and Dropout with L2 regularisation** – were assessed to identify their capability in protecting user images while maintaining model accuracy. The overall objective was to determine which methods provide the most balanced and effective privacy protection for AI image systems.

1.2.1 Objective 1 Conduct a literature review on image privacy leakage in AI platforms.

1.2.1.1 Gathered and reviewed peer-reviewed research papers discussing image privacy risk in AI platforms such as Gemini and ChatGPT, addressing both training and inference level vulnerabilities. [7 days]

1.2.1.2 Organised the selected literature into two categories- gradient leakage and model inversion attacks- representing the main source of image privacy threats. [5 days]

1.2.1.3 Summarised findings from each paper to highlight how user-upload images may be reconstructed or exposed during model processing. [3 days]

1.2.1.4 Deliverable(s): A structured literature review outlining privacy leakage issues, methodology relevant to AI platforms such as Gemini and ChatGPT.

1.2.2 Objective 2: Analyse gradient leakage attacks affecting AI image systems

1.2.2.1 Studied existing research on Deep Leakage from Gradients (DLG) and related attacks to understand how the gradients can expose private image data. [6 days]

1.2.2.2 Evaluated unbiased Gradient sampling (UGS) and GLAUS frameworks to determine their success and limitations in preventing gradient-based information leakage. [5 days]

1.2.2.3 Analysed experimental results from multiple studies to assess reconstruction quality and privacy risk.

1.2.3 Objective 3 – Analyse model inversion attacks in AI image platforms

1.2.3.1 Reviewed papers on generative model inversion (GMI) and other inversion-based attacks capable of reconstructing user images from model outputs. [6 days]

1.2.3.2 Examined how attackers use outputs, embedding, or confidence scores to recreate useruploaded images. [5days]

1.2.3.3 Compared white-box and black box-model inversion setups to identify privacy risks relevant to public AI image systems. [3days]

1.2.3.4 Deliverable(s): Comprehensive report summarising model inversion attacks, accuracy levels, and privacy implications.

1.2.4 Objective 4 – comparative and critical analysis

1.2.4.1 Integrated insights from gradient leakage and model inversion literature to understand combined privacy risk in AI image systems. [6days]

1.2.4.2 Compared modern defensive strategies such as CVB, dropout with L2 regularisation, and Data augmentation to evaluate their ability to reduce data Exposure without degrading performance. [4 days]

1.2.4.3 Identified research gaps and suggested improvements for enhancing image privacy protection in platforms like Gemini and ChatGPT. [3 days]

1.2.4.4 Identified research gaps and suggested improvements for enhancing image privacy protection in platforms like Gemini and ChatGPT. [3 days]

1.2.4.5 Deliverable(s): Comprehensive comparative report summarising effective privacy strategies and open research challenges.

1.2.5 Objective 5 – Review and evaluate defence mechanisms

1.2.5.1 Studied privacy defences including shadow Defense, BiDO, differential privacy (DP), Homomorphic Encryption, CVB, Dropout with L2 regulation, and Data augmentation to assess their effectiveness in preventing data leakage [6 days]

1.2.5.2 Extracted and compared performance metrics, and used to measure privacy protection and model accuracy. [5 days]

1.2.5.3 Identified advantages, limitations, and computational trade-offs of each defence approach. [4 days]

1.2.5.4 Deliverable(s): comparative summary of defences, highlighting which techniques best protect product image privacy in AI platforms.

1.2.6 Objective 6 – Documentation and presentation

1.2.6.1 Compiled all analysed results, comparisons, and evaluations into a complete academic report on AI image privacy leakage. [7 days]

1.2.6.2 Designed presentation slides summarising attacks, defences, and conclusions [3 days]

1.2.6.3 Finalised and submitted the full report for academic review. [2 days]

1.2.6.4 Deliverable(s): Final written report about privacy leakage mechanisms, defence results, and recommendations.

BACKGROUND RESEARCH

2.1 Attacks

Gradient Leakage

Gradient Leakage occurs when private training data can be reconstructed from shared gradient information during model training. Studies such as Yang et al. (2023) have shown that even when gradients are partially protected through techniques like unbiased gradient sampling (UGS), attackers can still reconstruct the data using advanced inversion algorithms such as GLAUS. These findings reveal that the gradient, commonly used in federated or distributed learning environments, can unintentionally expose the raw image.

Model inversion attacks

This attack works by studying a model's output instead of its training data. Attackers use the information that the model gives- such as output or internal feature to rebuild an image that looks like the one used during training. Zhang et al. (2020) introduced Generative model inversion (GMI), showing that attackers could recover faces and other details just from output confidence scores. For systems like Gemini and ChatGPT that deal with images, this could lead to personal data being revealed, even if the model itself is closed or protected.

These two attacks show that privacy risks exist even without direct access to user data. They are important to study because they show where and how sensitive information can leak in AI image systems.

2.2 Defences

Researchers have introduced several defence techniques to help reduce privacy risks in AI Image systems. Each method protects data at a different stage of the learning process, such as

during training, sharing, or prediction. The main goal of these defences is to make it harder for attackers to recover sensitive image data while still keeping the model accurate and efficient.

1. **Differential privacy:** Adds a small amount of random noise to model updates or outputs so that the individual images cannot be easily identified. This can sometimes slightly reduce the model accuracy.
2. **Homomorphic Encryption:** Keeps data encrypted while the model processes it. This means that even if the data is intercepted, it cannot be read in its original form. This can sometimes increase computational time.
3. **Convolutional variational bottlenecks:** introduce controlled randomness into the model's internal layer to make it harder for the attackers to reconstruct input images from the gradients. This method maintains a good balance between privacy and accuracy.
4. **Dropout with L2 Regularisation:** Prevents the model from memorising private details by randomly dropping some connections during training.
5. **Data augmentation:** Uses modified versions of the same images- such a color changes- to help the model learn general features instead of exact details, reducing the chance of memorising the sensitive data**.**
6. **Bilateral dependency optimisation:** Reduces the correlation between the learned features and the original input data, making it more difficult for attackers to reverse engineer the images.
7. **Shadow defence:** hides or changes sensitive gradient information before sharing it in distributed learning environments, reducing the risk of privacy leakage.
8. **Gradient clipping and quantisation:** Restricts and simplifies gradient information so that attackers cannot extract meaningful private data from it.

The use of these defence techniques is a key part of this project, as they help show how image privacy is strengthened in AI platforms such as Gemini and ChatGPT. By studying and comparing how effective each method it. The project aims to find which one offers the right balance between protecting user data and keeping the model accurate.

3. METHODOLOGY AND TECHNICAL APPROACH

This project uses a systematic literature-based research approach to study privacy leakage in AI image systems such as ChatGPT Gemini and other platforms. The work focuses on analysing and comparing

existing studies on two major privacy attacks- **Gradient Leakage and model inversion** – and several defensive methods designed to reduce image exposure.

Methodology

This research follows a comparative analytical process based on the principle of literature review. Reviewed papers and conference articles were collected from trusted databases, including IEEE Xplore, ScienceDirect, SpringerLink, and arXiv, focusing mainly on research published between 2020 and 2025 to ensure accuracy.

The reviewed literature was organised into two primary groups such as:

1. Attack-based studies, focusing on **Gradient leakage and model inversion.**
2. Defence-based studies, which include **Differential privacy, Homomorphic encryption, Convolutional variational Bottlenecks, Dropout with L2 regulation, data augmentation, bilateral optimisation, shadow defence, and gradient clipping**

This analysis helped to develop a clearer understanding of how image leakage occurs and how various defences are performed in practical applications.

All materials used in the study were obtained from public academic sources. No user data or AI system access was involved. The research complies with the **University of Greenwich's** ethical standards.

Technical Approach

The technical approach of the project is that each selected paper functions as part of an analytical testbed, representing experiments that have already been validated in prior research.

The process involves four key stages:

1. **Classification:** organising selected studies into attack-based and defence-based categories.
2. **Comparison:** Analysing the reported metrics, data sets, and results to understand privacy exposure levels and reconstruction success.
3. **Performance Assessment:** Evaluating how each defence method balances privacy protection, computational efficiency, and accuracy.
4. **Result integration:** Combining findings from all studies to determine which defence techniques offer the most practical and balanced protection for image data and AI platforms like ChatGPT and Gemini.

This approach ensures that the research remains structured, evidence-based, and academically rigorous, and it allows for meaningful comparison between different studies while concluding the best-performing strategies for reducing privacy leakage in AI image-generating platforms.

SYNOPSIS

Image-generative AI platforms such as Gemini and ChatGPT have made it easier to generate realistic images from users' text prompts. These tools are potent and creative, but they can also raise some serious

concerns about privacy and data leakage when users upload their own pictures to these kinds of AI platforms. One of the key issues is the **Gradient leakage** and the model inversion attacks, where the attacker can interrupt the gradient information shared when the model learning process. **Gradients are the mathematical vector representation shoes how much the model's internal weight should be adjusted to minimise the error when processing an image by user prompts**. The gradient works when the individual device calculates the gradient locally and sends it back to the central server for aggregation. The attacker can intercept these shared gradients, and he can mathematically reverse the process and reconstruct the actual raw image. In recent studies by Yang et al. (2023) it has been shown that even the advanced security frameworks in **federated learning**, such as **unbiased gradient sampling** (UGS), are still exposed to gradient leakage attacks (GLA). Their study introduced a technique called GLAUS, which demonstrated that attackers can accurately rebuild private images from gradient information – even when those gradients have been randomly sampled or transformed. These finding suggests that current aggregation and randomisation methods alone cannot fully protect user data. Recent research has introduced new ways to improve data privacy without heavily reducing accuracy. Scheliga, K., Amini, A., Althoff, T., and Kerzel, M. (2025) developed a method called **Convolutional Variational Bottlenecks (CVB)**, which adds small, controlled amounts of randomness during training. This makes it much harder for attackers to recover original images from gradients, while the model's accuracy stays consistent. Likewise, Ben Hamida, S., Ayed, S., and Atri, M. (2024) examined several defences, including **Differential Privacy, Dropout with L2 regularisation, and Data Augmentation**. Their findings showed that these techniques can effectively lower the risk of privacy breaches in machine learning systems.

Recent work has introduced advanced defences such as Shadow defences and bilateral Dependency Optimisation to further reduce the leakage. Shadow defence protects the sensitive region by adding targeted noise before gradients are shared, while the BIDO limits how much private image detail the model retains during training. Combined with **differential privacy, Dropout, and Data augmentation** can make it harder for attackers to reconstruct the users' raw images without major accuracy loss. **Differential privacy** protects the user data by adding a small amount of random noise to the information, such as gradients during training. This makes it impossible to tell whether any specific person or image data was used.

Dropout with L2 Regularisation During training, dropout randomly turned off some connections between neurons, which helps the model to avoid memorising private or unnecessary details from the user data. The **L2 Regularisation** part further reduces the influence of any single data point by slightly penalising large weight values. Together, they make the model more stable and less likely to leak the sensitive user data.

Data Augmentation helps protect privacy by making small changes to the training images, like flipping, rotating, or adjusting the lighting. These variations help the model learn general patterns

instead of remembering exact details from the specific picture. This way the AI can still learns effectively.

Keywords: [Gradient leakage, Model inversion Attack, Federated Learning, Data Privacy, Shadow Defence, Bilateral Dependency Optimisation, Differential Privacy, Convolutional Variational Bottleneck, Dropout, Data Augmentation, Image Generative AI, Privacy leakage]

a.4 Project Plan

1. Initial literature review

- Collected the main literature on privacy image leakage in the AI models
- One of the first key papers reviewed was Inverting Gradients: How easy is it to break privacy in Federated Learning
- This stage helped to define the project aim.

2. Review of defence strategies

- The reviewed defence methods are from the literature only at a theoretical level.
- Including the defence, Differential privacy, Dropout L2 regulation, and Variational bottlenecks.
- This stage showed that the project was becoming too theoretical without a practical demonstration.

3. Defence planning

- The first defence plan was to keep the project only theoretical. Later realised that was not enough to demonstrate the problem.
- Then tried to edit one of the defences from the literature review, selected Drouput L2 regulation. In this defence, the model automatically removes random neurons while training, the idea was to edit these defences to drop a particular side of the image while training due to less efficiency. The idea was dropped
- Then the defence was refined to Face pixelation during training.
- This was chosen because it was simpler and closely linked to the face de-identification literature, even though the defence only works in in certain trainings but the output was with less information of the facial structures.

4. Model planning and implementation

- The project moved from theoretical plan to a small proof of concept implementation.
- The main reason for building the model was the defence should not stay theoretical only.

- The baseline model made it possible to show the privacy problem more clearly and implemented a defence pipeline instead of stating the defence might work in the future.

5. Model choice

- The early plan was to use a large architecture like ResNet.
- Later, it was changed to CNN
- CNN was chosen because the CNN was more manageable and still suitable for the gradient leakage demonstration.

6. Development environment

- Google Colab was chosen as the implementation environment.
- It was simple to use, easy to access, and supported the Python-based projects.

7. Timing of the revision

- The final structure of the project was done by weeks 14-15
- At this stage, the project becomes more focused and moves fully in the baseline vs the defended comparison.

8. The final stage

- Testing the prototype
- Comparing outputs using metrics
- Writing the final report
- Preparing posters for the Demonstration

APPENDIX B - Testing documentation

b.1 Testing plan

The testing in this project focused on the following points.

- checking that the baseline pipeline could reconstruct an approximate image from leaked gradients
- checking that the defended pipeline could also run successfully after the defence was applied
- comparing the baseline and defended outputs visually
- recording image quality metrics, including MSE, PSNR, and SSIM
- seeing whether the defended reconstruction looked less clear and less revealing than the baseline reconstruction
- repeating the process to see whether the same overall pattern remained, even if the exact output changed slightly between runs.

b.2 Baseline pipeline Testing

The baseline pipeline was tested to confirm that the prototype could reconstruct an approximate image from leaked gradients. The output showed visible facial structure and general visual information, which confirmed that the leakage process worked as expected. The reconstruction was then evaluated using MSE, PSNR, and SSIM.



Fig 3.1 reconstructed baseline with metrics

b.3 Defended pipeline testing

The defended pipeline was tested using the same overall process, but with the defence applied before training. The output image appeared more blurred and less detailed than the baseline case. This suggested that the defence reduced how much useful visual information could be reconstructed from the gradients. The defended output was also evaluated using MSE, PSNR, and SSIM.

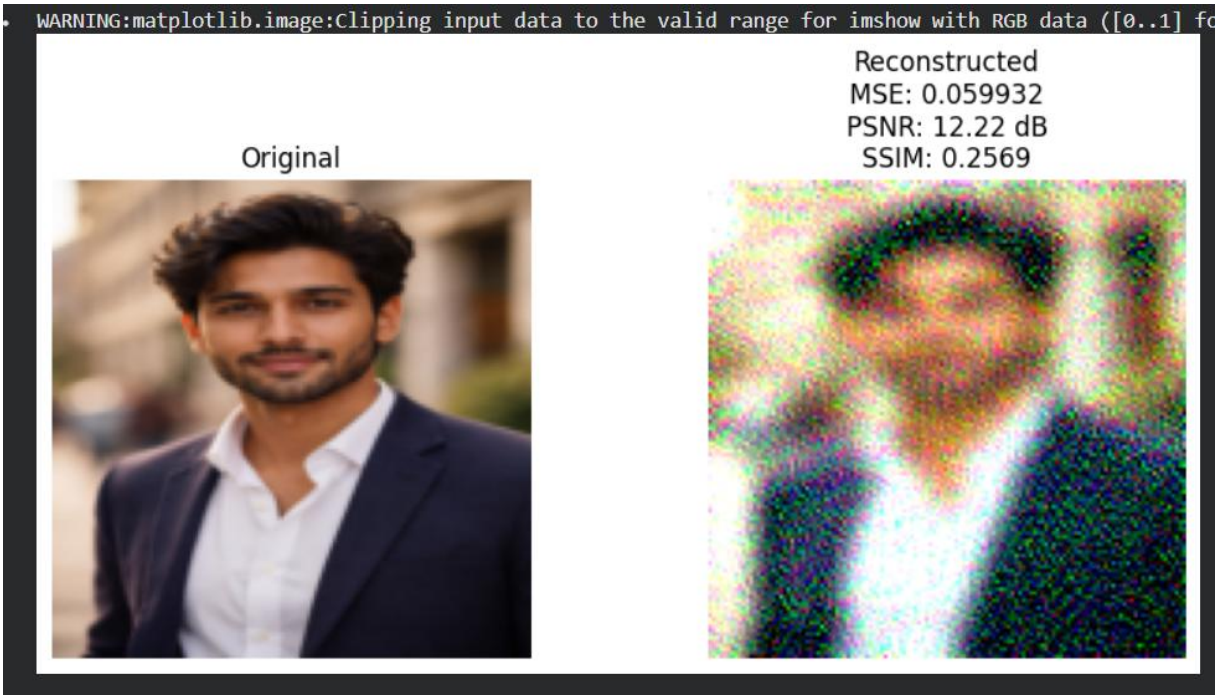


Fig 3.2 defended pipeline with metrics

b.4 Repeated testing result

Metric	Baseline result	Defended result
MSE	0.042849	0.059932
PSNR	13.68 db	12.22 db
SSIM	0.4148	0.2569

b.5 Repeated testing result

Repeated runs showed that the exact reconstruction and metric values changed slightly between experiments. This was expected because gradient inversion is an optimisation-based process, and small differences in gradient matching can affect the result. However, the overall pattern remained the same, with the baseline output appearing clearer than the defended output.

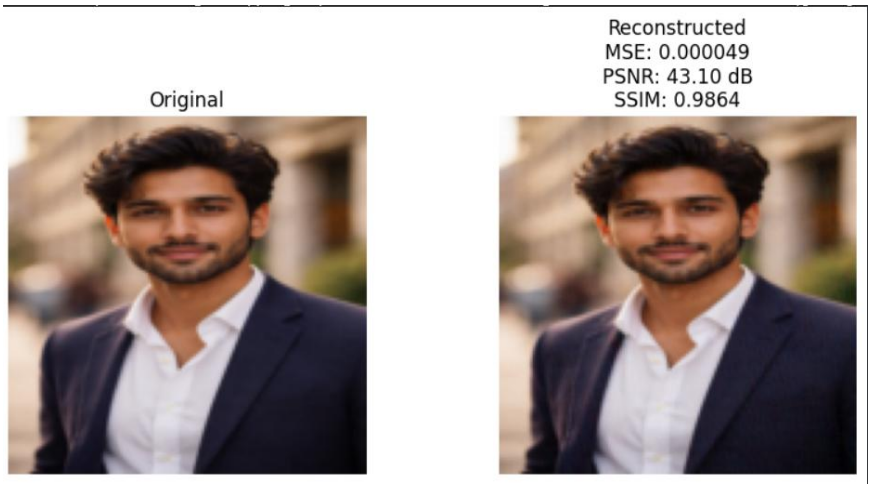


Fig 3.3 shows the changes between runs

This image shows the change in the output for the baseline in the second run, the reconstruction was more accurate.

b.6 Supporting literature comparison

For context, earlier studies have shown gradient inversion attacks on larger architectures such as ResNet. An example from the literature is included below to show that existing research has explored more advanced settings than the smaller CNN-based proof-of-concept used in this project.

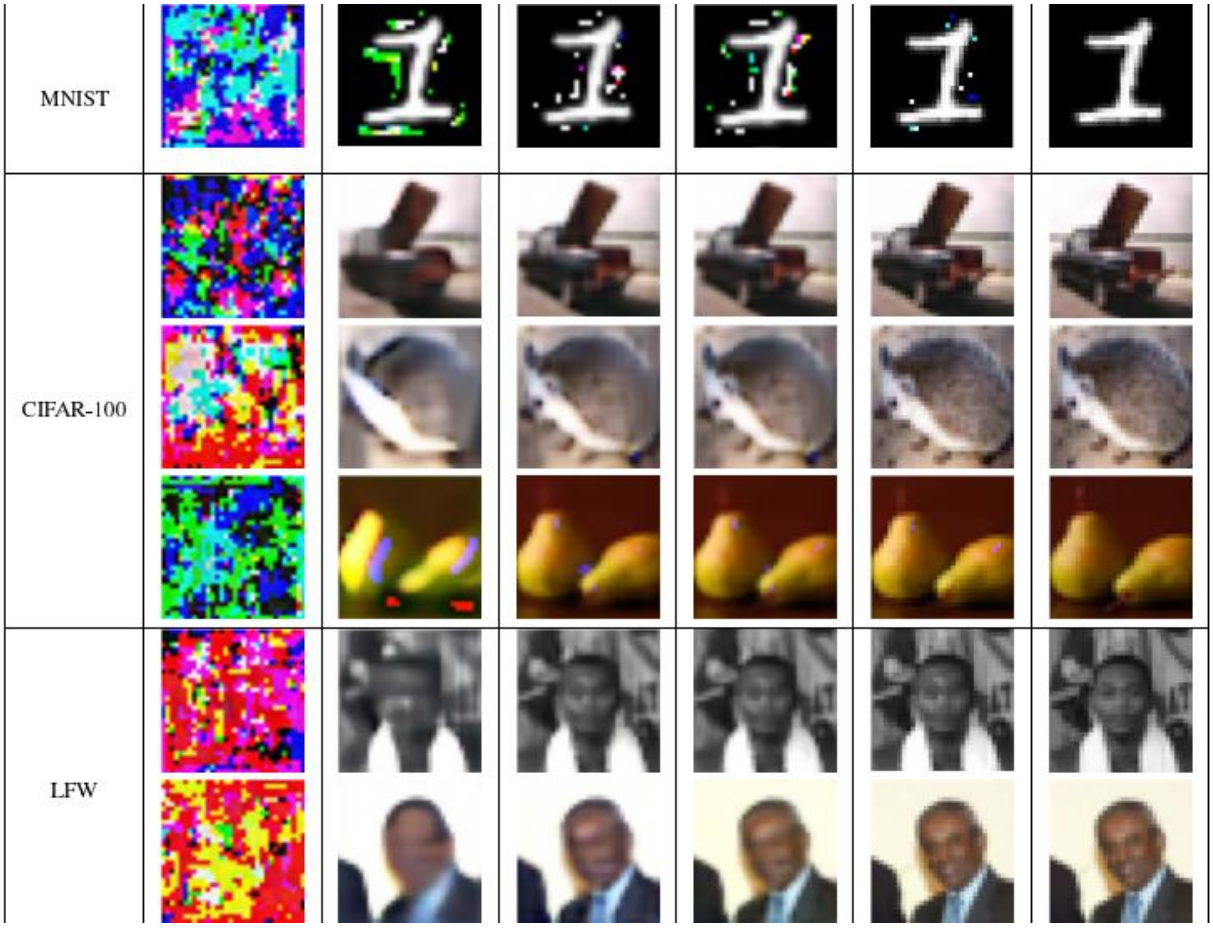


Fig 3.4 Reconstruction using the ResNet from the literature, Inverting Gradients: how easy is it to break privacy in Federated learning.